CLASS 8 MATHEMATICS :: SLIP TEST – FA-4

Questions

Question 1.

Subtract $(3a - 4b + 5c)$ from $(7a - 2b - 3c)$.

Question 2.

Multiply the binomials $(2x + 3)$ and $(3x - 4)$.

Question 3.

Using the standard identity $(a - b)^2 = a^2 - 2ab + b^2$, evaluate $(2x - 5y)^2$.

Question 4.

Factorise the algebraic expression completely by grouping terms: $am + bm + an + bn$.

Question 5.

Factorise the quadratic expression: $x^2 + 7x + 12$.

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Solutions

Solution 1:

$= (7a - 2b - 3c) - (3a - 4b + 5c)$

$= 7a - 2b - 3c - 3a + 4b - 5c$

$= (7a - 3a) + (-2b + 4b) + (-3c - 5c)$

$= \mathbf{4a + 2b - 8c}$

Solution 2:

$= (2x + 3)(3x - 4)$

$= 2x(3x - 4) + 3(3x - 4)$

$= 6x^2 - 8x + 9x - 12$

$= \mathbf{6x^2 + x - 12}$

Solution 3:

$(a - b)^2 = a^2 - 2ab + b^2$

Here $a = 2x$, $b = 5y$

$(2x - 5y)^2 = (2x)^2 - 2(2x)(5y) + (5y)^2$

$= \mathbf{4x^2 - 20xy + 25y^2}$

Solution 4:

$am + bm + an + bn$

$= (am + bm) + (an + bn)$

$= m(a + b) + n(a + b)$

$= \mathbf{(a + b)(m + n)}$

Solution 5:

$x^2 + 7x + 12$

$(3 + 4 = 7,\ 3 \times 4 = 12)$

$= x^2 + 3x + 4x + 12$

$= x(x + 3) + 4(x + 3)$

$= \mathbf{(x + 3)(x + 4)}$